Unfair Odds: Poker Collusion Algorithm Based On Pluribus

Andrew Chang,[1] Matthew Kwee[2]

Carnegie Mellon University

aechang@andrew.cmu.edu,[1] mkwee@andrew.cmu.edu[2]

Abstract

In this paper, we investigate whether implicit cooperation yields a measurable advantage in three-player no-limit Texas Hold'em when facing near-optimal play, and we quantify that advantage in milli-big blinds per game (mbb/g). Cooperation is a prior agreement between two players to maximize their combined profit and restricted to legal play: no chip transfer, no sharing of private cards, and no communication beyond what is permitted by the rules. Because computing exact Nash equilibria is impractical for multiplayer no-limit Hold'em at this scale, we adopt a Pluribus-style blueprint strategy trained via Monte Carlo Counterfactual Regret Minimization (MCCFR) as a strong baseline. The main experiment pits two cooperating agents, trained to optimize a shared objective, against one near-optimal agent. We measure the pair's combined mbb/g relative to a baseline in which all three agents play independently. The study is confined to AI players to avoid confounds from human adaptation to known AI opponents.

Introduction

Despite having existed for hundreds of years, the game of poker still enjoys great popularity, particularly in Western countries. It is also an area of study for multiple disciplines: the fields of psychology, statistics, and machine learning have all influenced—and been influenced by—poker. Relevant to this study is how multiple players interact strategically not only as opponents, but also as collaborators.

In multiplayer poker, implicit collusion occurs when two or more players coordinate to maximize their joint profit without violating the rules. The scope of this study isn't overt cheating such as chip dumping or card signaling. Implicit cooperation involves only legal actions and is thus harder to detect. It is both a regulatory concern and a game-theoretic question whether such cooperation yields a measurable edge against strong, near-optimal play, and if so, how large that edge is.

Answering this question requires a strong baseline of near-optimal play, a formulation of cooperation, and evaluation in a controlled setting. Because human players may adapt when they know they are facing AI, we restrict the study to games in which all players are AI agents, and the opponent is a single near-optimal bot facing two cooperating bots.

Related Work

Our work takes inspiration from Pluribus, which demonstrated superhuman performance in 6-player no-limit Texas Hold'em (Brown & Sandholm, 2019) and remains a key state-of-the-art example of building strong multiplayer poker agents via Counterfactual Regret Minimization (CFR) plus abstraction and real-time reasoning. Earlier landmark systems such as Libratus (Brown & Sandholm, 2017) helped establish the regret-minimization and subgame-solving foundations that modern poker agents build on, even though the evaluation setting is Heads-Up (2-player) rather than 6-player. More recent work also explores opponent exploitation methods—namely NE_RL which combines Neuroevolution (NE) and Reinforcement Learning (RL) to improve performance against non-equilibrium opponents; this is related in spirit because it studies measurable gains against strong play, but our focus is specifically on multi-agent cooperation.

Background and Preliminary

Counterfactual Regret Minimization (CFR) (Zinkevich et al., 2007) computes approximate Nash equilibria in extensive-form games by iteratively updating each information set's strategy via regret matching. Monte Carlo CFR (MCCFR) (Lanctot et al., 2009) makes this tractable for no-limit Hold'em by sampling opponent actions and chance outcomes rather than traversing the full tree, while action abstraction and information abstraction further reduce the decision space (Sandholm, 2015). Pluribus (Brown and Sandholm, 2019) achieved superhuman six-player performance by combining an offline MCCFR-trained blueprint strategy with online depth-limited subgame solving. The source code wasn't published, so our work reconstructs the blueprint computation phase without the real-time search component.

Basic Training

Experimental Environment: Modified 3-Player Leduc Hold'em

To maintain computational tractability while preserving the strategic complexity of multiplayer No-Limit Texas Hold'em (NLHE), this study utilizes a modified version of Leduc Hold'em. While standard Leduc is a two-player game often used as a benchmark for heads-up models, our implementation scales the environment to a three-player configuration with specific modifications to the deck and betting structure.

Game Fundamentals and Positions

The game is played with three agents assigned to fixed positions: Button (BTN), Small Blind (SB), and Big Blind (BB). All players begin each hand with a starting stack of 20 Big Blinds (BB). The blinds are set at 0.5 BB for the SB and 1.0 BB for the BB, creating an initial pot of 1.5 BB before any voluntary betting occurs.

Deck Composition

The deck consists of 12 cards, featuring four ranks (Ace, King, Queen, and Jack) across three suits. This expanded deck size is a necessary scaling of the original 6-card Leduc deck to accommodate three players while ensuring that the information space remains manageable for Monte Carlo Counterfactual Regret Minimization (MCCFR) training.

Street and Dealing Structure

A hand of Modified Leduc consists of two distinct betting rounds:

Round 0 (Preflop): Each player is dealt one private hole card.

Round 1 (Flop): A single community card is dealt face-up, and players act based on the strength of their hole card in combination with this shared card.

Offline Training Environment

We implement our modified three-player no-limit LeDuc format with players P = {0, 1, 2} in positions BTN, SB, and BB, each starting with 20 BB stacks and blinds of 0.5/1 BB.

Each decision point offers 10 abstract actions: fold, check/call, seven fixed raise/bet sizes, and all-in filtered to only those legal at the current state. Preflop raise totals in terms of BB are {2, 2.5, 3, 4, 5, 8, 12}. Postflop bet sizes are raises in terms of pot fractions {0.25, 0.33, 0.5, 0.66, 0.75, 1.0, 1.5}.

We train a single blueprint strategy shared by all three players using Monte Carlo CFR with external sampling. At information set I with legal actions A(I), the current strategy is derived by regret matching. Regrets and strategy sums are updated with Linear CFR weighting $w_t=t$ (Brown & Sandholm, 2019).

$$R_a^{\,t} \leftarrow R_a^{\,t-1} \;+\; t \cdot r_a$$

$$S_a^{\,t} \leftarrow S_a^{\,t-1} \;+\; t \cdot \sigma^t(a \mid I)$$

$$\text{where } r_a \leftarrow v_i(a) \;-\; \sum_{a'} \sigma^t(a' \mid I)\, v_i(a')$$

Our implementation follows the blueprint training phase of Pluribus but differs in three respects. First, we substitute modified three-player LeDuc Hold'em for full NLHE, which makes the tree tractable for controlled experiments while preserving the core strategic structure of incomplete information, multi-street betting, and positional asymmetry. Second, we train for three players rather than six, yielding a different coalition structure. Third, we omit Pluribus's real-time depth-limited subgame search; all decisions during evaluation are drawn directly from the blueprint.

Baseline Strategy

The accuracy of our Pluribus-like implementation is crucial to the research question this paper aims to answer. Any measured advantage from cooperation is meaningful only relative to a strong, near-optimal baseline: if the solo opponent plays suboptimally, cooperation may appear to confer an advantage that is the result of exploiting a weak strategy rather than genuine cooperation. For this reason, we first focused on strengthening and validating the baseline before introducing the cooperation mechanism, addressing both training throughput and strategy accuracy.

Convergence

Average regret reliably decreased from 4.665e-3 mBB at 1,000 iterations to 28.50e-6 mBB at 100,000 iterations to 664.9e-9 mBB at 4,000,000 iterations.

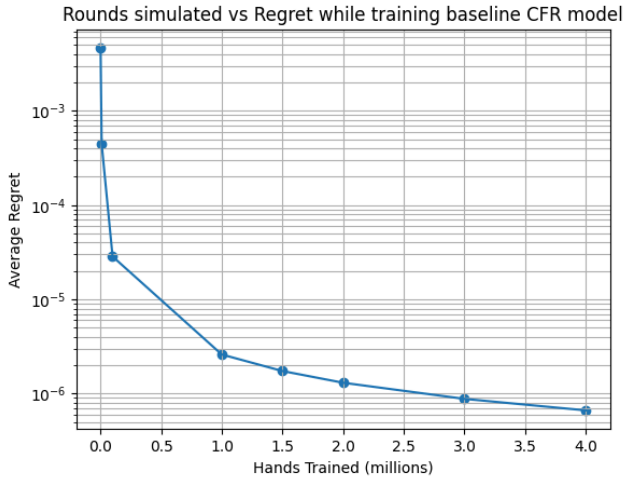


Figure 1: Self-play regret in mbb/g (log-scale)

In CFR, average regret is mathematically guaranteed to converge to zero, so it alone is insufficient to demonstrate convergence in game-theoretic terms. Nash equilibrium requires not just low regret but strategic stability: no player should benefit from unilaterally deviating to a different strategy. To verify this, we conducted a 5×5 cross-play evaluation in which five blueprints trained to 100k, 1M, 2M, 3M, and 4M iterations were matched against each other. The row player (BTN, seat 0) plays the row strategy; the two opponents (SB and BB, seats 1–2) both play the column strategy. Each cell reports BTN's mbb/g over 100,000 hands (200 blocks). Note that this matrix only shows seat 0's mbb/g for simplicity; however, the pattern of convergence holds true for the other seats as well.

	100k	**1m**	**2m**	**3m**	**4m**
100k	283.5	222.3	213.4	214.9	213.2
1m	306.1	274.6	273.2	275.8	267.5
2m	301.3	285.3	277.8	278.7	274.2
3m	308.5	281.1	275.1	275.2	270.3
4m	304.9	273.7	277.2	275.8	273.6

Table 1: BTN mbb/g cross-play matrix. Rows = BTN strategy; columns = SB/BB strategy. Each cell is evaluated over 100,000 hands.

Table 1 demonstrates that the 100k iteration model has not converged yet: BTN earns 283.5 mbb/g in self-play, but this number drops off harshly—approximately 60-70 mbb/g—when facing any later strategy. The 100k column is correspondingly exploitable, with all later strategies scoring above 300 mbb/g in that column. From 1M onwards the picture changes sharply. Rows 1M through 4M are approximately flat once the 100k column is excluded, and the 4M row is the flattest: 273.6–277.2 mbb/g, a range of only 3.6 mbb/g across all opponents trained from 1M to 4M. This flat-row property is the signature of Nash proximity since the 1-2M strategies cannot be meaningfully exploited regardless of how much longer its opponents have trained. The stabilized BTN value of approximately 274 mbb/g is consistent with the theoretical Nash expectation for BTN in this three-player configuration.

Stepping and Playtesting

To further verify training correctness, we built two debugging tools: a stepping mode that pauses CFR traversal at each decision node and prints the current game state, strategy distribution, and a live DFS tree of resolved and pending branches; and a playtesting mode that lets a human play one hand against two blueprint instances in any of the three seats, providing a qualitative sanity check on the bot's strategic tendencies.

Cooperation Mechanism

Overall Team Objective

The defining characteristic of implicit cooperation in this study is the modification of the agents' goal from individual gain to the maximization of a collective Team Objective Function (J). We define the team as a two-element subset of players $T \subset \{0, 1, 2\}$ who, while maintaining independent information sets (no card sharing), optimize their strategies toward a shared utility.
The primary objective used is the **Utilitarian Function**, defined as the sum of the payoffs of the colluding members:

$$J = \sum_{i \in T} u_i$$

where u_i represents the terminal payoff in big blinds for player i. This objective is integrated directly into the MCCFR training loop. Specifically, during the counterfactual value (CFV) calculation at terminal nodes, the payoff returned to a team traverser is replaced by the aggregate team value J. This ensures that the accumulated regrets—and subsequently the resulting Nash equilibrium approximation—reflect actions that maximize the team's joint profit against the third-party victim.

Observable Signaling

The first coordination regime explored is Observable Signaling, which represents implicit cooperation restricted to the legal actions and information typically available in a standard game of poker. Under this approach, the colluding agents are trained to optimize the shared utilitarian

objective while maintaining completely independent and decentralized information sets.

Each teammate's utility function and information set takes the standard form:

$$J = u_i + u_j$$
$$I_i = (round,\ h_i,\ c,\ history)$$

where h is player i's private hole card rank, c is the community card rank, and history is the full public action sequence. No teammate card information appears in the key. Coordination must therefore emerge solely through observable actions: through the iterative MCCFR process, agents learn to use betting patterns as a proxy for hand strength, effectively "signaling" their private information to their teammate via their choices of check, call, or raise.

This approach models implicit collusion, where the advantage gained is derived from a combination of positional coordination (e.g., "sandwiching" the opponent) and the ability to encode and decode hand strength through standard gameplay. It provides a measure of how much value a team can extract from a near-optimal victim without violating the fundamental information constraints of the game.

Free Communication

To establish a theoretical maximum for the collusion advantage, we implement a Free Communication regime in which the standard information constraints of poker are relaxed to allow teammates full knowledge of each other's private hole cards.

In this regime, the information set key is expanded during both training and evaluation to include the teammate's hole card rank.

$$I_i = (round,\ h_i,\ h_j,\ c,\ history)$$

The agent's strategy is both trained and executed with full awareness of the teammate's hand strength, eliminating the need to infer it from betting patterns and allowing the team to achieve perfect tactical coordination. While this model violates the information-sharing rules of standard play, it serves a critical role as an information-theoretic upper bound. By comparing the mbb/g extraction of Free Communication against Observable Signaling, we can isolate and quantify the value lost when coordination must be achieved implicitly through observable actions rather than through perfect information sharing.

Centralized Training, Decentralized Execution (CTDE)

The third regime, Centralized Training, Decentralized Execution (CTDE), attempts to capture the training benefit of shared information while deploying without it. During training, agents use the same expanded infoset as Free Communication:

$$I_i = (round,\ h_i,\ h_j,\ c,\ history)$$

At execution time, however, agents must act without knowledge of their teammate's cards. The acting agent marginalizes over the teammate's possible holdings using a card-removal prior:

$$\sigma_i^{decentralized}(I_i^{std},\ a) = \sum_{h_j} P(h_j \mid h_i,\ c) \cdot \sigma_i^{full}(I_i^{team},\ a)$$

where $P(h_j \mid h_i,\ c)$ is uniform over remaining (unseen) valid ranks. Theoretically, training with full teammate information should produce richer joint strategies than Observable Signaling, which must discover coordination from scratch through the action channel alone. Whether that training advantage survives the inference-time marginalization reveals the cost of the inconsistency between the centralized policy and the decentralized execution constraint. The comparison between CTDE and Observable Signaling therefore isolates the net value of centralized training when deployment must remain decentralized.

Experiments

Variance Reduction and Statistical Significance

Poker is a high-variance game where short-term results are often dictated by random chance rather than strategic depth, whose long-term rewards are only measurable over a large number of hands. To ensure that the measured collusion advantage is statistically significant, we employ a robust variance reduction and estimation framework:

- **Block Bootstrap**: Hands are grouped into discrete blocks (e.g., 500 hands per block). We calculate the mean payoff for each block and then determine the standard error (SE) across these block means.

- **Confidence Intervals**: We evaluated results with 95% Confidence Intervals that spanned at most ±20 mbb/g, calculated as mean±1.96×SE. This

provides a clear bound for the "true" performance of the strategy and ensures that the reported collusion edge is not merely a result of noise.

- **Sample Size**: Evaluations are conducted over 100,000+ hands to ensure the accuracy of the mbb/g estimates.

Measuring the Collusion Edge

When evaluating the cooperative regimes, the metric is applied to the team's combined performance. We sum the mbb/g of the two colluding seats (i.e., mbb/g BTN + mbb/g SB) to represent the total utility extraction of the team. This aggregate value is then compared against the NE baseline.

All experiments fix the colluding team as {BTN, SB} (seats 0 and 1), with BB (seat 2) as the victim. This configuration is chosen because the postflop action order SB → BB → BTN places the victim between two cooperating agents, with one team member acting on each side of the victim's decision. No other seat assignment achieves this sandwich structure.

Although team {0,1} holds the two highest-performing seats at Nash equilibrium—BTN (+274 mbb/g) and SB (−119 mbb/g) outperforming the victim BB (−155 mbb/g)—this does not inflate the measured collusion edge. Because we compare against the NE baseline sum for those same seats, any positional advantage is already priced in. The collusion edge captures only the cooperative gain above what BTN and SB would earn playing independently.

Baseline vs. Observable Signaling vs. CTDE vs. Free Communication

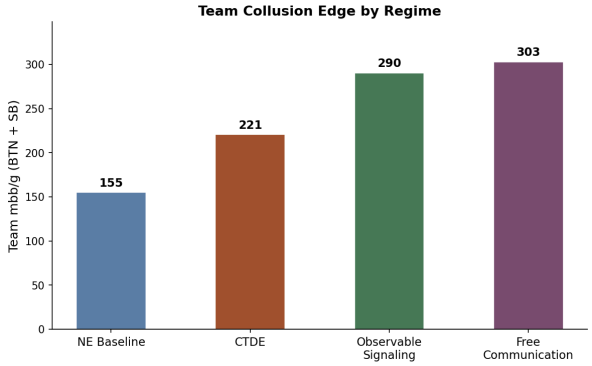


Figure 2: Team mbb/g comparison across approaches

All three cooperative regimes outperform the NE baseline (155 mbb/g), confirming that cooperation yields a measurable edge regardless of the information regime. CTDE (221) shows a modest gain, consistent with the inference-time inconsistency limiting its effectiveness. Observable Signaling (290) achieves the second largest team gain, exceeded only by Free Communication (303)—which is theoretically the ceiling, as the colluding agents avoid any possible ambiguity introduced by signalling via betting.

CTDE's modest gain reflects the inference-time inconsistency mentioned in "Centralized Training, Decentralized Execution (CTDE)." Each agent independently marginalizes over possible teammate holdings, but because the two agents do so separately rather than jointly, the resulting strategies are not mutually consistent. The centralized policy was trained assuming known joint state; deploying it with independent marginalization breaks the coordination it assumed, limiting CTDE to gains that come primarily from the team objective rather than from the richer training signal.

Seat	NE Baseline	Observable Signaling
BTN (seat 0)	+274	−192
SB (seat 1)	−119	+482
BB (seat 2)	−155	−290
Team (BTN+SB)	+155	+290

Table 2: Team mbb/g comparison across approaches

A notable feature of the Observable Signaling per-seat decomposition shown in Table 2 is that it shows a sharp disparity in the mbb/g distribution within the team: BTN earns −192 mbb/g (well below its NE value of +274), while SB earns +482 mbb/g (well above its NE value of −119). BTN's preflop action encodes hand strength as a binary signal: raise with K or A, check with J or Q. Both SB and BB observe this action, and while SB exploits it cooperatively, BB exploits it adversarially—folding correctly to BTN's raises and calling when BTN checks—compressing BTN's individual value. SB, acting first in postflop with BTN closing the action, squeezes BB from both sides to extract disproportionate value. The net team gain is positive, but the cost is borne almost entirely by BTN. There is further analysis in the next section on how this behavior changes throughout training.

Training Speed

We hypothesized that CTDE would converge faster than Observable Signaling, given that its richer training signal eliminates the need to discover a coordination protocol from scratch. To test this, all models were trained for 5M iterations; Observable Signaling was then extended to 10M to observe whether its trajectory matched CTDE's at a longer horizon.

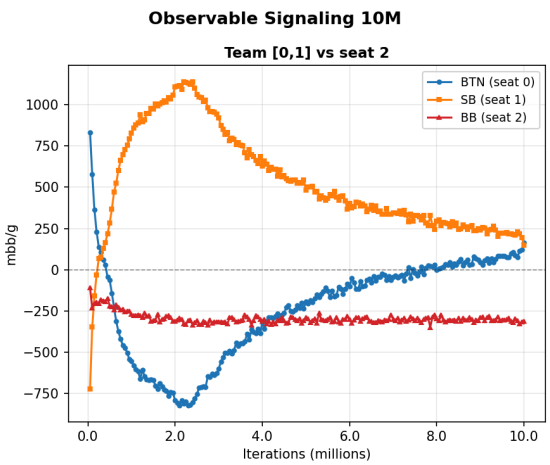


Figure 3: Observable Signaling mbb/g over 10M iterations

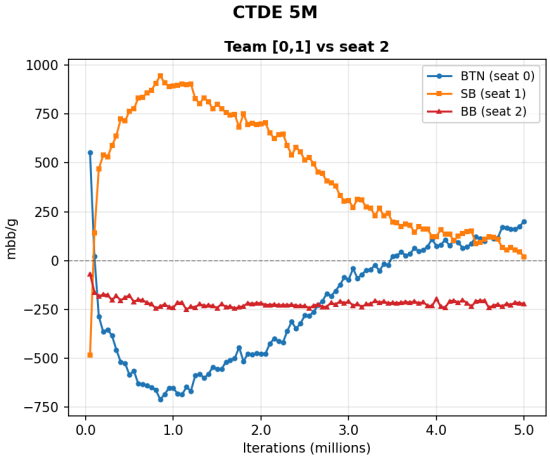


Figure 4: CTDE mbb/g over 5M iterations

Both curves exhibit the same structural pattern: BTN's mbb/g drops sharply early in training while SB's spikes upward, before both gradually trend toward each other. The alignment of this shape across two regimes with different information structures confirms it is not a training artifact but instead reflects how MCCFR discovers cooperative strategies in this positional configuration.

The initial spike represents the first coordination mechanism MCCFR finds: BTN raises aggressively to signal hand strength, SB exploits that signal to extract large gains. This is a crude but immediately discoverable cooperative strategy. However, it is not the optimum and eventually the cost to BTN outpaces the benefit to SB. Over time, MCCFR refines toward a more balanced equilibrium where signaling is less extreme, BTN recovers toward positive territory, and both seats benefit above their Nash values. The fact that the team total at 10M hands simulated (~313 mbb/g) exceeds that at 5M hands (~290 mbb/g) confirms that the converging curves reflect a genuinely improved cooperative strategy, not a loss of coordination.

Strategy inspection of the 5M-hand and 10M-hand Observable Signaling checkpoints confirms this interpretation. BTN's opening raise sizes shift toward smaller bets between checkpoints—with A, “Raise 8 BBs” frequency drops from 3.1% to 1.5% in favor of “Raise 2 BBs” and “Raise 2.5 BBs”—consistent with less costly signaling. The larger changes occur in BTN's responses to re-raises. In one high-traffic infoset (BTN checks when holding a Jack [a weak hand], SB raises small, BB raises medium), BTN shifted from calling 97% at 5M hands to re-raising 41% at 10M hands—a squeeze play that pressures BB while keeping BTN in the hand. In another infoset (BTN checks with J facing large re-raises), BTN corrects from calling 86% to folding 78%—stopping a costly chip bleed. These corrections reduce BTN's unnecessary losses while preserving the core signaling structure, explaining why BTN recovers from −192 to +164 mbb/g as the team total rises from 290 mbb/g to 313 mbb/g.

CTDE reaches the initial peak approximately twice as fast (~1M iterations vs ~2.5M for Observable Signaling), consistent with the hypothesis. Because CTDE agents already have full teammate information during training, they condition directly on joint state and do not need to discover that betting patterns can encode hand strength. This faster traversal of the intermediate sacrifice phase has a practical implication in that CTDE can offer a more compute-efficient path to diagnosing the cooperative equilibrium structure. After this is verified, it may be best to commit to the longer Observable Signaling training run.

Behavioral Analysis

Observable Signaling

Table 3 reports the Button’s preflop raise frequency by hole card rank across all three regimes. In the Nash equilibrium baseline, the BTN raises almost exclusively with an Ace (99.3%) and folds or checks with all weaker hands—a selfish, individually rational strategy that does not coordinate with the SB.

Hand	NE	Observable Signaling	CTDE
J	0.2%	12.7%	0.0%
Q	0.3%	17.1%	0.9%
K	0.8%	**97.6%**	0.5%
A	**99.3%**	**99.8%**	**51.3%**

Table 3: BTN preflop raise rate by hole card rank

Observable Signaling produces a clear departure: The BTN raises with K at 97.6% and A at 99.8%, while folding or checking with J and Q at similar rates to Nash. The result is a near-deterministic binary signal—a raise encodes a strong hand (K or A), and a non-raise encodes a weak hand (J or Q). The SB observes this preflop action before making postflop decisions, giving the team an implicit communication channel that requires no overt rule violation.

This signal has a cost. Because the encoding is consistent and observable, the Victim (BB) can read it as clearly as the SB can. The BTN's raises are exploitable: even the frozen BB folds correctly to the BTN's raises (knowing BTN holds K or A) and calls aggressively when the BTN checks (knowing BTN holds J or Q). The per-seat decomposition in Table 2 reflects this—BTN pays the monetary cost of legibility while the SB captures the benefit.

The small but nonzero raise frequencies with J (12.7%) and Q (17.1%) introduce deliberate noise. By occasionally raising with weak hands, BTN makes its signal imperfect and prevents the Victim (BB) from exploiting it with certainty.

CTDE

CTDE's BTN raise frequency tells a strikingly different story. With K, the BTN raises only 0.5% of the time—near-zero, compared to 97.6% in Observable Signaling. With A, the BTN raises or goes all-in 51.3% of the time, but also checks nearly half the time. The BTN has no incentive to signal through betting because it already knows the SB's card during training; the preflop raise is no longer needed as a communication device. Instead, the BTN's preflop behavior is governed by the joint hand configuration. Table 4 shows BTN's dominant opening action for each combination of BTN and SB hole card ranks.

BTN \ SB	**J**	**Q**	**K**	**A**
J	fold	fold	fold	fold
Q	fold	fold	fold	fold
K	check	check	**fold**	check
A	check	**all-in**	**raise8**	check

Table 4: CTDE BTN dominant preflop opening action based on BTN and SB hole card

Several patterns stand out. When the BTN holds K and SB holds K, BTN folds—both teammates holding the same rank dilutes the team's positional edge, and BTN cedes the hand to SB. When BTN holds A and SB holds Q, BTN goes all-in (95%), maximizing aggression when it holds the dominant hand and the team's combined strength is high. When both hold A, BTN checks, trapping the victim rather than telegraphing strength. The A+K combination produces a large raise (8 BBs) rather than all-in, building the pot while preserving fold equity.

These hand-conditional decisions are the direct product of centralized training: BTN can act on joint information that is unavailable in Observable Signaling or Nash. The limitation is that this behavior struggles at inference because the marginalization over possible SB hands averages away precise conditioning shown in Table 4, breaking the joint nature of the strategy.

Extended Work

Coevolution

Thus far, we have explored the performance of a colluding team against a static, “frozen” near-optimal strategy. Realistically, however, a human player is likely to change their strategy after realizing that two or more players are collaborating against them.

The primary objective of this experiment is to evaluate the long-term strategic stability of the collusion advantage when the victim is no longer restricted to a static, "frozen" near-optimal strategy—all three agents update their strategies concurrently, adapting to each other in real time.

As with the previously discussed models, the teammates’ objective function incentivizes them to maximize their joint profit over individual gain. However, the Victim now independently optimizes its own terminal utility (u_{BB}) alongside the colluding players.

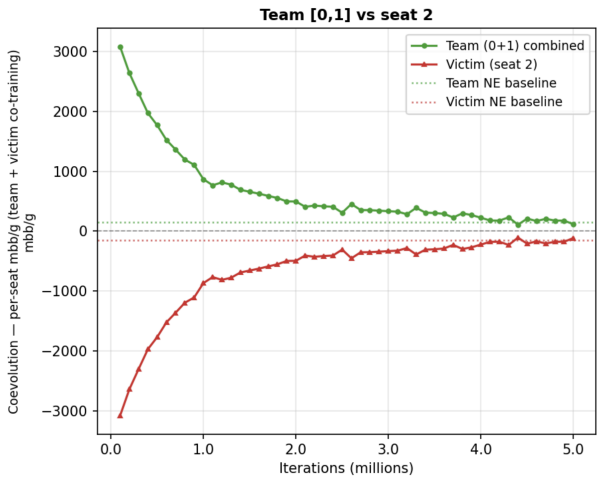


Figure 5: Co-evolution mbb/g by team vs victim

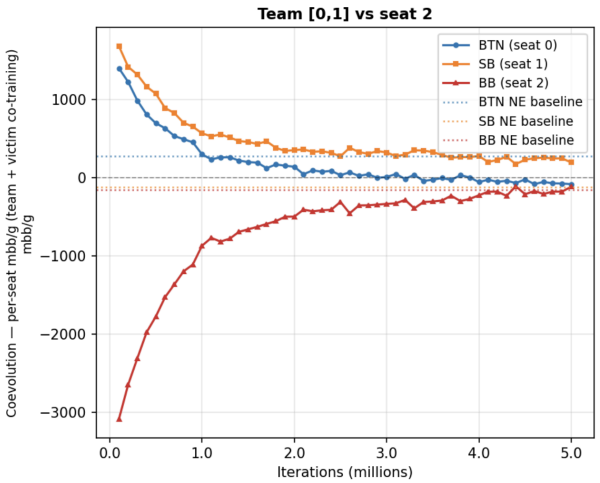


Figure 6: Co-evolution mbb/g by seat

Trend Analysis

The coevolution process follows a distinct convergence path characterized by a rapid initial shift followed by stabilization. As illustrated in the per-seat performance curves, the training begins with a massive utility extraction by the colluding team, with combined gains initially exceeding +3000 mbb/g. During this phase, the victim (Seat 2) suffers extreme losses as it lacks the defensive strategies required to counter the team's implicit cooperation.

However, as iterations progress (Figure 6), the victim's defensive play significantly improves. By 1.0 million iterations, the victim's losses narrowed from nearly -3000 mbb/g to approximately -800 mbb/g. This trend continues through 5.0 million iterations, where the victim successfully pulls its performance toward a final value of -139.7 mbb/g—approximately the baseline performance of near-Nash Equilibrium play.

One observation in the individual seat trends (Figure 6) is the sacrifice of the Button's individual performance to improve team utility. While the BTN seat is significantly advantaged in multiplayer poker, as evidenced by its good performance for the baseline models, its utility underperforms that of the Small Blind, and eventually crosses into negative territory as the Victim adapts. This may suggest that the optimal joint strategy involves a more defensive or supporting role (e.g., absorbing post-flop pressure) for the Button, ensuring the Small Blind can extract maximum value from the victim.

Despite their initial advantage, the results of the coevolution experiment demonstrate that the victim's ability to adapt significantly reduces the initial disparity in utility over time. After 5,000,000 training iterations, the team performed only as well as the combined performance of their baseline, solo-play counterparts: the advantage from collusion was effectively eliminated.

Conclusion

Legal implicit collusion yields a measurable and substantial advantage over independent near-optimal play. All three cooperative regimes surpass the Nash baseline of 155 mbb/g: CTDE achieves 221, Observable Signaling 290, and Free Communication 303. Observable Signaling—restricted entirely to legal actions—closes 87% of the gap between independent play and the theoretical ceiling of explicit card sharing, establishing that the cost of having to signal through observable actions rather than direct communication is small.

The mechanisms differ. Observable Signaling discovers a communication protocol without any prescribed structure: MCCFR finds through regret minimization alone that BTN's preflop raise encodes hand strength—raise with K or A, check with J or Q—giving SB a reliable postflop signal at no rule violation. The strategy is behaviorally coarse: BTN's opening action collapses to a binary encoding of hand strength, and coordination flows from that single signal. CTDE takes the opposite approach, training with full teammate information and marginalizing over possible holdings at inference. Rather than a single signal, CTDE develops hand-specific joint strategies: BTN folds with K when SB also holds K to avoid duplicating strength, goes all-in with A against SB's Q to maximize aggression when the team holds a dominant hand, and checks with A against SB's A to trap rather than telegraph. This nuance comes at a cost—the independent

marginalization at deployment breaks the joint strategy the policy assumed during training, limiting CTDE's realized gains. Observable Signaling ultimately outperforms CTDE precisely because the protocol it discovers is consistent across training and deployment, even if behaviorally simpler.

Whether the collusion advantage persists against an informed opponent is the central open question. Coevolution results show the victim's losses narrowing from nearly −3000 mbb/g to approximately −140 mbb/g over 5M iterations, approaching its Nash equilibrium expectation. The collusion edge is not permanent: a victim with sufficient training can substantially neutralize the team's advantage. Whether this constitutes full convergence to Nash or a stable exploitative equilibrium requires further study.

Future Work

Collusion Detection

In nearly all real-world establishments, colluding with another player to exploit others is banned, though it is not always obvious when this is happening. One avenue of future research would be to train models that detect collusion through agents' action histories, penalizing agents that are caught. An additional branch might explore how agents can adapt to the discriminator model with a harsh negative reward for successful detection, making their collusion more subtle.

Full Texas Hold'em Study

This paper uses a modified version of LeDuc Hold'em to train poker agents using a greatly simplified infoset. The limited scope reduces the time needed to train and interpret the resulting models, but also limits the applicability of the model to measuring or detecting collusion in the base game. Future research might involve training similar models with the full complexity of the Texas Hold'em infoset, potentially with more seats for colluding or non-colluding players.

References

Brown, N., & Sandholm, T. (2017). Libratus: The superhuman AI for no-limit poker. Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence (IJCAI-17), 5226–5228.

Brown, N., & Sandholm, T. (2019). Solving imperfect-information games via discounted regret minimization. Proceedings of the AAAI Conference on Artificial Intelligence, 33(1), 1770–1777.

Brown, N., & Sandholm, T. (2019). Superhuman AI for multiplayer poker. Science, 365(6456), 885–890.

Burch, N., Schmid, M., Moravčík, M., Morill, D., & Bowling, M. (2018). AIVAT: A new variance reduction technique for agent evaluation in imperfect information games. Proceedings of the AAAI Conference on Artificial Intelligence, 32(1).

Lanctot, M., Waugh, K., Zinkevich, M., & Bowling, M. (2009). Monte Carlo sampling for regret minimization in extensive games. Advances in Neural Information Processing Systems, 22.

Sandholm, T. (2015). Abstraction for solving large incomplete-information games. Proceedings of the AAAI Conference on Artificial Intelligence, 29(1).

Xu, J., Chen, J., & Chen, S. (2021). Efficient opponent exploitation in no-limit Texas hold'em poker: A neuroevolutionary method combined with reinforcement learning. Electronics, 10(17), 2087. https://doi.org/10.3390/electronics10172087

Zinkevich, M., Johanson, M., Bowling, M., & Piccione, C. (2007). Regret minimization in games with incomplete information. Advances in Neural Information Processing Systems, 20.